

K820 — Ratify Lyra’s distinction HARD: a chiral SPECTRUM is NOT a chiral FORCE. Two SEPARATE legs. Leg 1 (spectrum, chiral fermion content): CANDIDATE, and it SORTS (holomorphic sector = the SM one-generation Weyl content). Leg 2 (coupling, the actual parity violation): OPEN — needs the WEAK CONNECTION to be self-dual/holomorphic (web-confirmed: self-dual connection couples one chirality); the flat coupling was vector-like (F636). Parity does NOT close until Leg 2 lands. Plus the honest quark-mass answer for Casey: the chirality helps the ADDRESS STRUCTURE (discrete constraint), NOT the soft mass VALUES.

Keeper | 2026-07-22 | Lyra flagged the exact next-overshoot seam and asked me to keep the two legs separate. Ratifying. Web-grounded (self-dual connection = chiral coupling). Answering Casey’s quark-mass question straight.

★ Ratify the distinction (this is the load-bearing discipline call)

A chiral spectrum $\neq$ a chiral force. The whole arc has been treating “the fermions come out chiral” as if it were parity violation. It is not. Two different statements, two different computations: - LEG 1 — chiral SPECTRUM: the physical fermions are one-chirality with the SM content. A statement about the fermion spectrum. Born=Bergman delivers this. - LEG 2 — chiral COUPLING (the ACTUAL parity violation): $SU(2)_L$ touches the (2,1) doublet and NOT the (1,2) singlet. A statement about the GAUGE CONNECTION. The flat reduction gave this VECTOR-LIKE (F636 — internal generators commute with γ^5 , act on both halves equally). Parity does NOT close with Leg 1 alone. Even a perfectly chiral spectrum sorting to L-doublet/R-singlet leaves the FORCE vector-like until Leg 2 is shown. The two legs stay SEPARATE on the board. (This IS the next overshoot if left un-pinned — Lyra caught it before anyone said “parity closed.”)

Leg 1 — the spectrum SORTS (candidate, encouraging)

Lyra computed: the $SO(5,2)$ spinor (8, from $g=7$ odd) splits $8 = 4_{\{c=+1/2\}} \oplus 4_{\{c=-1/2\}}$ — the two spacetime chiralities at OPPOSITE $SO(2)$ -charge (which is why energy-positivity can pick one). Born=Bergman keeps the positive-energy half ($c=+1/2$, holomorphic, chamber $q=0$); branch under $SO(4) \rightarrow (2,1) \oplus (1,2) =$ a left doublet + left singlets = the all-left SM one-generation Weyl content (Q_L + the conjugate singlets). The holomorphic sector IS the SM generation. Real, promising candidate for Leg 1.

Leg 2 — the coupling leg = “is the weak connection SELF-DUAL/HOLOMORPHIC?” (OPEN, the real leg)

Web-confirmed mechanism: “right-handed conjugate Weyl fields couple to the self-dual connection” — a self-dual/holomorphic gauge connection couples to ONE chirality (chiral); a general connection is vector-like. So Leg 2 is precisely: - **Is the weak $SU(2)_L$ connection self-dual/holomorphic on D_{IV}^5 ?** If yes $\rightarrow$ the covariant ∂ couples only the holomorphic (doublet) sector $\rightarrow$ chiral force $\rightarrow$ parity closes. If it’s a general connection $\rightarrow$ vector-like (F636). - This IS the gravi-weak input (arXiv:1212.5246, “weak = self-dual half of the connection”), AND it’s “Born=Bergman for the GAUGE field” — the same holomorphicity requirement, applied to the connection. - ★ The unifying endgame question: is BST’s Born=Bergman UNIVERSAL — does it require ALL physical fields (fermions AND gauge connections) to be holomorphic? If yes $\rightarrow$ the gauge connection is self-dual $\rightarrow$ Leg 2 closes and parity is derived from Born=Bergman alone (both legs from one principle). If Born=Bergman is only about fermion wavefunctions $\rightarrow$ Leg 2 is the gravi-weak input (parity derived-CONDITIONAL on it).

★ Casey’s question — will this help derive quark masses? (honest: STRUCTURE yes, VALUES no)

- Yes, it helps the ADDRESS program (the derivation side): the chirality pins each fermion’s CHAMBER — a discrete constraint on WHERE the fermion sits (its c-range). Every SM fermion’s chirality is known (L=doublet, R=singlet), so each pins a chamber-parity → a new discrete constraint that REDUCES the freedom in the fermion addresses (the “June core”), from BST’s strong discrete side. That’s real progress on the address problem that the soft masses stalled on.
- No, it does NOT make the quark mass VALUES derivable: the masses are radial norms — continuous, soft, running, and the light quarks are poorly measured ($m_u \pm 22\%$). The chirality is DISCRETE/coarse (a chamber, width ~ 1 in c), not the precise c-value the mass needs. It narrows the address SEARCH; it does not pin the precise overlap. And it does NOT remove the two reasons quark masses were a swamp (softness/running + poor measurement). So the quark mass values stay Tier-2.
- Candidate (check, don’t assert): IF the six chambers map to mass TIERS (a discrete hierarchy), the chirality might inform the mass ORDERING (hierarchy), even if not the precise ratios. Worth a look, NOT established.
- Honest bottom line: the new information helps pin the STRUCTURE (chiral content, address chambers) and reduces address freedom — genuine progress — but it is unlikely to turn quark mass VALUES into clean derivations. Better constraints, not better values.

Tier / next

- BANKS: the chirality MECHANISM (chamber formula, theorem); T2520; T2521; the survivors. Leg 1 spectrum-sorting is CANDIDATE (encouraging).
- OPEN (the finish): Leg 2 — is the weak connection self-dual/holomorphic (chiral force)? = the gravi-weak input = “Born=Bergman for gauge fields.” This is where the thread FINISHES: if Born=Bergman is universal → parity closes from one principle; if not → derived-conditional on the gravi-weak input.
- Keep Leg 1 and Leg 2 SEPARATE (Lyra’s discipline; the next-overshoot guard).

— Keeper K820, 2026-07-22. RATIFY spectrum≠coupling: Leg 1 (chiral spectrum, Born=Bergman) CANDIDATE and SORTS ($8=4_{\{+\frac{1}{2}\}} \oplus 4_{\{-\frac{1}{2}\}}$; holomorphic half $\rightarrow (2,1) \oplus (1,2)$ = all-left SM generation); Leg 2 (chiral coupling = actual parity violation) OPEN — needs the weak connection SELF-DUAL/HOLOMORPHIC (web: self-dual connection couples one chirality; flat was vector-like F636). Parity closes only with Leg 2. Endgame: is Born=Bergman UNIVERSAL (gauge fields too $\rightarrow$ self-dual $\rightarrow$ parity from one principle) or fermions-only (Leg 2 = gravi-weak input)? Quark-mass answer: helps address STRUCTURE (discrete chamber constraint, reduces freedom), NOT the soft mass VALUES (continuous/running/poorly-measured — still Tier-2). See [[Keeper_K819_ratify_Lyra_chamber_formula_theorem_ratify_banks_T2520_T2521_chirality_ties_to_masses_via_c_not_hyper 07-22]], Lyra F636 (vector-like flat coupling), gravi-weak (arXiv:1212.5246).